

Shuyu (William) Zhang

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EDUCATION

University of Pennsylvania — M.S.E. in Mechanical Engineering

Expected May 2027

Relevant Coursework (*In Progress*): CIS 520 Machine Learning (ML algorithms + generalization theory), CIS 580 Machine Perception

Southwest Jiaotong University – University of Leeds Joint School — B.E. in Mechanical Engineering

June 2025

GPA: 3.54 / 4.00

TECHNICAL SKILLS

Languages: Python (primary), C, JavaScript/TypeScript (reading proficiency)

ML / Robotics: PyTorch, SE(3) geometric control, A* path planning, inverse kinematics

Infrastructure: Docker, PostgreSQL, Cloudflare Tunnel, Linux, Git

Web (architected via Claude Code): Next.js, React, TypeScript, full-stack deployment

AI & Agent Tools: Claude Code + OpenClaw — architecture-level familiarity; MCP, skills, hooks, memory, user/project scope

PROJECTS

Volunteer Tracker — Full-Stack Volunteer Management Platform GitHub | Live

Jan – Apr 2026

Solo project · Built with Claude Code · Next.js / TypeScript · Prisma / PostgreSQL · Docker · Cloudflare Tunnel

- **Shipped** a production volunteer management platform for Puregold Classic Translation, serving 100+ active volunteers. Deployed on a personal Mac mini behind Cloudflare Tunnel (live).
- **Owned** all architecture and product decisions while pairing with Claude Code as the implementation agent. Core features: account system with 4-tier role-based access control, volunteer filtering on an interactive Leaflet map, service-record submission (self and proxy, with owner confirmation), project-level management, and an aggregated ledger view.
- **Drove** continuous evolution — schema versions v2.1 → v3.1 → v3.2 and a mid-project backend migration from Express to Next.js — paired with automated pg_dump GFS backups (7 daily / 4 weekly / 12 monthly), keeping every iteration data-safe.
- **Designed and operates a multi-layer self-healing ops loop:** OpenClaw (local agent) runs scheduled cron health checks and auto-repairs, while an SSH channel from my dev machine handles complex interventions — the two channels **repair each other** (OpenClaw maintains the SSH tunnel; SSH recovers OpenClaw's state when it hangs). UptimeRobot serves as the last-resort external alerting layer.

Autonomous Quadrotor Navigation — MEAM 6200 Advanced Robotics

Jan – Mar 2026

Course project · Python · Hardware: Bitcraze CrazyFlie

- Implemented SE(3) geometric controller (Lee et al.) for stable 3D flight in simulation.
- Built 3D path planner with A* search and greedy line-of-sight pruning; applied minimum-jerk polynomial trajectory generator.
- Achieved collision-free flight through 3 obstacle mazes on real hardware (sim-to-real transfer).

Vision-Guided Robotic Manipulation — MEAM 5200 Intro to Robotics

Nov – Dec 2025

Course project · Python · ROS · Gazebo · Hardware: Franka Emika Panda

- Built autonomous pick-and-place pipeline (perception → planning → execution) on a 7-DOF Franka Panda arm, handling both static objects and a dynamically rotating platform.
- Implemented inverse kinematics solver and motion planner; developed in Ubuntu / ROS / Gazebo, deployed to real hardware addressing sim-to-real calibration gaps.

Additional: Hybrid Autonomous & WiFi-Controlled Competition Robot (MEAM 5100, team of 4, Nov-Dec 2025) — led mechanical design (Ackermann chassis, SolidWorks); supported embedded + software (ESP32-C3, dual-mode autonomous + WiFi). See LinkedIn.

AI TOOLS & AGENT ENGINEERING

- **Authored 3 Claude Code articles on Rednote** (572+ likes/saves) on platform *internals and harness integration* — most-viewed: “Wiring MiniMax M2 into the CC harness”; others cover CC's native task/memory primitives and operational modes.
- **Practices an opinionated agent-steering philosophy:** teach the agent to understand the need and the underlying mechanism rather than forbid actions or mandate behavior — treat it as a partner, not a worker.